Statistics for AI/ML-1

A Complete Foundation

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1. Why Statistics for AI/ML?

Machine learning is, at its core, applied statistics. A model is a function that estimates patterns from data, and almost every step — collecting data, training, evaluating, deploying — is a statistical operation:

- Data is a **sample** drawn from an unknown distribution.
- Training is **parameter estimation**.
- Loss functions are derived from **likelihood** principles.
- Generalization is a statement about **sampling and bias**.
- Evaluation metrics rely on **hypothesis testing** and **confidence intervals**.
- Regularization, dropout, ensembles — all reduce **variance**.

Statistics gives the vocabulary and machinery to reason about uncertainty, which is the central problem of AI.

2. Types of Data and Variables

Before any analysis, identify what kind of data you have. The wrong assumption breaks everything downstream.

2.1. Qualitative vs. Quantitative

- **Qualitative (Categorical)**: labels with no inherent numerical meaning. Examples: blood type, country, product category.
- **Quantitative (Numerical)**: numbers that meaningfully measure something.

2.2. Levels of Measurement

- **Nominal**: categories without order (e.g., colors).
- **Ordinal**: ordered categories without uniform spacing (e.g., movie ratings: bad, ok, good).
- **Interval**: numeric with uniform spacing but no true zero (e.g., temperature in °C).
- **Ratio**: numeric with true zero (e.g., height, weight, count).

2.3. Discrete vs. Continuous

A **discrete** variable takes countable values (number of clicks). A **continuous** variable takes uncountably many values within a range (image pixel intensities are technically discrete but treated as continuous).

Use case in AI/ML

The level of measurement decides feature engineering. Nominal features need one-hot or embedding encoding; ordinal features can be label-encoded; ratio/interval features are usually standardized. Tree-based models tolerate raw categories better than linear models, which assume numeric structure.

3. Descriptive Statistics

Descriptive statistics summarize the structure of a dataset before any modeling.

3.1. Measures of Central Tendency

3.1.1. Mean

The arithmetic average of n values:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

The expected value $\mathbb{E}[X]$ is the population analog.

3.1.2. Median

The middle value when data is sorted. Robust to outliers — the median income of a country is more meaningful than the mean if one billionaire lives there.

3.1.3. Mode

The most frequent value. Useful for categorical data and multimodal distributions.

Intuition

If the mean and median differ substantially, your data is likely skewed. Skew means linear models (which optimize mean squared error and therefore chase the mean) may produce biased predictions.

Use case in AI/ML

Imputation: Missing values in tabular data are commonly filled with the mean (numeric), median (numeric with outliers), or mode (categorical). **Baseline models**: "predict the mean" or "predict the majority class" are the floors any real model must beat.

3.2. Measures of Dispersion

3.2.1. Range

Range $= \max(x) - \min(x)$. Sensitive to outliers; rarely used alone.

3.2.2. Variance

The average squared deviation from the mean:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 \quad \text{(population)}, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \quad \text{(sample, Bessel's correction)}.$$

The $n-1$ denominator (Bessel's correction) makes s^2 an unbiased estimator of σ^2.

3.2.3. Standard Deviation

$\sigma = \sqrt{\sigma^2}$. Same units as the data, so it's interpretable.

3.2.4. Quartiles, IQR, and Percentiles

- **Q1, Q2 (median), Q3**: split data into four equal parts.
- **IQR** (Interquartile Range) $= Q3 - Q1$: a robust spread measure.
- **Outlier rule of thumb**: values outside $[Q1 - 1.5 \cdot \text{IQR}, Q3 + 1.5 \cdot \text{IQR}]$.

Use case in AI/ML

Feature scaling: `StandardScaler` ($\frac{x-\mu}{\sigma}$) requires unit-free, comparable scales for gradient-based optimizers, k-NN, SVM, and PCA. **RobustScaler** uses median and IQR for outlier-heavy features. **Outlier detection**: IQR rules feed into data cleaning pipelines.

3.3. Shape: Skewness and Kurtosis

3.3.1. Skewness

Measures asymmetry:

$$\gamma_1 = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right].$$

Positive skew: long right tail (income); negative skew: long left tail (exam scores ceiling-capped).

3.3.2. Kurtosis

Measures tail heaviness:

$$\gamma_2 = \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^4\right] - 3.$$

Excess kurtosis > 0 means heavier tails than Gaussian (more extreme values). Stock returns famously have heavy tails.

Use case in AI/ML

Many ML algorithms (linear regression, Gaussian Naive Bayes, PCA) assume approximately Gaussian features. Strong skew motivates transforms: $\log(1+x)$, Box–Cox, or Yeo–Johnson. Heavy-tailed targets often warrant Huber loss instead of MSE.

3.4. Covariance and Correlation

3.4.1. Covariance

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] \,.$$

Sign tells direction; magnitude depends on units.

3.4.2. Pearson Correlation

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

Measures *linear* association. $\rho = 0$ does *not* imply independence.

3.4.3. Spearman and Kendall

Rank-based correlations that capture monotonic (not just linear) relationships and are robust to outliers.

Use case in AI/ML

Feature selection: drop one of any pair of highly correlated features ($|\rho| > 0.9$) to reduce multicollinearity in linear models. **EDA**: correlation heatmaps reveal target leakage and redundant variables. **PCA** is fundamentally an eigendecomposition of the covariance matrix.

4. Probability Theory

Probability is the formal language for uncertainty. Every ML model output ultimately expresses uncertainty about an unobserved variable.

4.1. Foundations

4.1.1. Sample Space, Events, Axioms

A **sample space** Ω is the set of possible outcomes. An **event** $A \subseteq \Omega$. Kolmogorov's axioms:

(i) $P(A) \geq 0$ for all events A.

(ii) $P(\Omega) = 1$.

(iii) For disjoint $A_1, A_2, \ldots$: $P(\bigcup_i A_i) = \sum_i P(A_i)$.

4.1.2. Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

4.1.3. Independence

A and B are independent iff $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$.

4.1.4. Law of Total Probability

For a partition $\{B_i\}$ of Ω:

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

4.2. Bayes' Theorem

$$P(H \mid D) = \frac{P(D \mid H)\,P(H)}{P(D)}.$$

$P(H)$ is the prior, $P(D \mid H)$ the likelihood, $P(D)$ the evidence, $P(H \mid D)$ the posterior.

Intuition

Bayes' rule tells you how to update beliefs in light of evidence. It is the engine of the entire Bayesian school of ML.

Use case in AI/ML

Naive Bayes classifiers (spam filters, document classification) directly apply Bayes' rule under a conditional independence assumption. **Bayesian neural networks** maintain a posterior over weights. **Probabilistic programming** (Stan, Pyro, PyMC) is built on it. **A/B testing** under a Bayesian framework continuously updates the posterior over treatment effect.

5. Random Variables

A **random variable** X is a function from Ω to $\mathbb{R}$. It assigns numbers to outcomes so we can do math on them.

5.1. Discrete Random Variables

Defined by a **probability mass function (PMF)**: $p(x) = P(X = x)$, with $\sum_x p(x) = 1$.

5.2. Continuous Random Variables

Defined by a **probability density function (PDF)** $f(x)$ where

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Note: $f(x)$ is a density, not a probability — it can exceed 1.

5.3. Cumulative Distribution Function (CDF)

$$F(x) = P(X \le x).$$

Always monotonic, right-continuous, with $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.

5.4. Expectation, Variance, Moments

$$\mathbb{E}[X] = \sum_x x\,p(x) \quad \text{or} \quad \int x f(x)\,dx.$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Linearity of expectation (holds regardless of independence):

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

Variance is not linear:

$$\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

5.5. Joint, Marginal, and Conditional Distributions

For two RVs:

- **Joint**: $p(x, y)$ or $f(x, y)$.
- **Marginal**: $p(x) = \sum_y p(x, y)$ or $f(x) = \int f(x, y)\,dy$.
- **Conditional**: $p(y \mid x) = p(x, y)/p(x)$.

Use case in AI/ML

ML models learn conditional distributions $p(y \mid x)$ from samples of joint $p(x, y)$. Generative models (VAEs, diffusion, GANs) learn the joint or marginal $p(x)$ itself. The chain rule $p(x, y) = p(y \mid x)p(x)$ underlies sequence models like RNNs and Transformers, where $p(x_1, \ldots, x_T) = \prod_t p(x_t \mid x_{<t})$.

6. Important Probability Distributions

6.1. Discrete Distributions

6.1.1. Bernoulli

A Bernoulli random variable represents a single experiment with only two possible outcomes: Success (1) or Failure (0).

- **Support:** $x \in \{0, 1\}$
- **Parameter:** $p = P(X = 1)$
- **PMF:**
$$P(X = x) = p^x(1 - p)^{1-x}$$
- **Mean:**
$$E(X) = p$$
- **Variance:**
$$Var(X) = p(1 - p)$$

Use case in AI/ML

A Bernoulli distribution models binary outcomes such as:

- Spam / Not Spam
- Fraud / Not Fraud
- Disease / No Disease
- Click / No Click

In machine learning, the target variable in binary classification is often assumed to follow a Bernoulli distribution. Logistic Regression predicts the Bernoulli probability $p = P(Y = 1|X)$.

6.1.2. Binomial

Binomial Distribution

A Binomial random variable counts the number of successes in n independent Bernoulli trials.

- **Support:**
$$x = 0, 1, 2, \ldots, n$$
- **Parameters:**
$$n = \text{number of trials}, \quad p = \text{success probability}$$
- **PMF:**
$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$
- **Mean:**
$$E(X) = np$$
- **Variance:**
$$Var(X) = np(1-p)$$

Use case in AI/ML

The Binomial distribution models the count of successful outcomes:

- Number of customers who click an advertisement
- Number of defective items in a batch
- Number of students passing an exam

In machine learning, Binomial models are used when predicting counts of successes out of a fixed number of trials, such as conversion rates and A/B testing experiments.

6.1.3. Poisson

A Poisson random variable counts the number of events occurring within a fixed interval of time, space, or volume.

- **Support:**
$$x = 0, 1, 2, \ldots$$
- **Parameter:**
$$\lambda = \text{average event rate}$$

- **PMF:**
$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$
- **Mean:**
$$E(X) = \lambda$$
- **Variance:**
$$Var(X) = \lambda$$

Use case in AI/ML

The Poisson distribution models event counts:
- Website visits per minute
- Customer arrivals per hour
- Insurance claims per month
- System failures per day

In machine learning, Poisson Regression is used to predict count-valued targets such as clicks, purchases, incidents, and demand forecasting.

6.1.4. Geometric

A Geometric random variable represents the number of trials required until the first success occurs.

- **Support:**
$$x = 1, 2, 3, \ldots$$
- **Parameter:**
$$p = \text{success probability}$$
- **PMF:**
$$P(X = x) = (1-p)^{x-1}p$$
- **Mean:**
$$E(X) = \frac{1}{p}$$
- **Variance:**
$$Var(X) = \frac{1-p}{p^2}$$

Use case in AI/ML

The Geometric distribution models waiting time until the first success:
- Number of calls until a sale occurs
- Number of website visits until a purchase
- Number of emails sent until a reply

In machine learning and business analytics, Geometric models help estimate expected waiting time for a desired event such as conversion, response, or system success.

Distribution	What It Models
Bernoulli	One success/failure experiment
Binomial	Number of successes in n trials
Poisson	Number of events in an interval
Geometric	Trials until first success

6.2. Continuous Distributions

6.2.1. Uniform

A Uniform random variable assigns equal probability density to every value in an interval.

- **Support:**
$$a \leq x \leq b$$
- **Parameters:**
$$a = \text{lower bound}, \quad b = \text{upper bound}$$
- **PDF:**
$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$
- **Mean:**
$$E(X) = \frac{a+b}{2}$$
- **Variance:**
$$Var(X) = \frac{(b-a)^2}{12}$$

Use case in AI/ML

The Uniform distribution models complete uncertainty where every value is equally likely. Examples:

- Random number generation
- Random initialization of neural network weights
- Simulation and Monte Carlo methods

Uniform distributions are often used as prior distributions and for random sampling in machine learning.

6.2.2. Gaussian (Normal)

The Normal (Gaussian) distribution is the most important continuous distribution in statistics and machine learning.

- **Support:**
$$-\infty < x < \infty$$
- **Parameters:**
$$\mu = \text{mean}, \quad \sigma^2 = \text{variance}$$
- **PDF:**
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- **Mean:**
$$E(X) = \mu$$

- **Variance:**
$$Var(X) = \sigma^2$$

Use case in AI/ML

Many real-world quantities approximately follow a Normal distribution:
- Heights and weights
- Measurement errors
- Sensor noise
- Model residuals

In machine learning:
- Linear Regression assumes normally distributed errors.
- Gaussian Naive Bayes assumes Normal features.
- PCA relies heavily on Gaussian assumptions.

6.2.3. Exponential

The Exponential distribution models the waiting time until the next event occurs.

- **Support:**
$$x \geq 0$$

- **Parameter:**
$$\lambda = \text{event rate}$$

- **PDF:**
$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- **Mean:**
$$E(X) = \frac{1}{\lambda}$$

- **Variance:**
$$Var(X) = \frac{1}{\lambda^2}$$

Use case in AI/ML

The Exponential distribution models waiting times:
- Time until the next customer arrives
- Time until machine failure
- Time until the next website click

It is the continuous analogue of the Geometric distribution and is fundamental in survival analysis and reliability engineering.

6.2.4. Beta

The Beta distribution models probabilities and proportions.

- **Support:**
$$0 \leq x \leq 1$$
- **Parameters:**
$$\alpha > 0, \quad \beta > 0$$
- **PDF:**
$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$
- **Mean:**
$$E(X) = \frac{\alpha}{\alpha + \beta}$$
- **Variance:**
$$Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Use case in AI/ML

The Beta distribution models uncertainty about probabilities:
- Conversion rate
- Click-through rate
- Probability of success

In Bayesian machine learning, the Beta distribution is the conjugate prior for the Bernoulli and Binomial distributions.

6.2.5. Gamma

The Gamma distribution models the waiting time until multiple events occur.

- **Support:**
$$x \geq 0$$
- **Parameters:**
$$\alpha = \text{shape}, \quad \beta = \text{rate}$$
- **PDF:**
$$f(x) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$
- **Mean:**
$$E(X) = \frac{\alpha}{\beta}$$
- **Variance:**
$$Var(X) = \frac{\alpha}{\beta^2}$$

Use case in AI/ML

The Gamma distribution generalizes the Exponential distribution.
Examples:

- Waiting time until the 5th customer arrives
- Total rainfall
- Insurance claim sizes

Gamma distributions are commonly used in Bayesian statistics and stochastic process modeling.

6.2.6. Student's t

The Student's t distribution is similar to the Normal distribution but has heavier tails.

- **Support:**
$$-\infty < x < \infty$$
- **Parameter:**
$$\nu = \text{degrees of freedom}$$
- **PDF:**
$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$
- **Mean:**
$$E(X) = 0, \quad \nu > 1$$
- **Variance:**
$$Var(X) = \frac{\nu}{\nu - 2}, \quad \nu > 2$$

Use case in AI/ML

The Student's t distribution is used when:

- Sample size is small
- Population variance is unknown

Applications:

- t-tests
- Confidence intervals
- Bayesian regression

As $\nu \to \infty$, the t distribution approaches the Normal distribution.

6.2.7. Chi-squared

The Chi-Square distribution is obtained by summing squared standard Normal variables.

- **Support:**
$$x \geq 0$$
- **Parameter:**
$$k = \text{degrees of freedom}$$

- **PDF:**
$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$$
- **Mean:**
$$E(X) = k$$
- **Variance:**
$$Var(X) = 2k$$

Use case in AI/ML

The Chi-Square distribution is widely used in:
- Goodness-of-fit tests
- Independence tests
- Variance estimation

Many statistical hypothesis tests in machine learning rely on Chi-Square distributions.

6.2.8. F-distribution

The F distribution is the ratio of two independent Chi-Square random variables divided by their degrees of freedom.

- **Support:**
$$x > 0$$
- **Parameters:**
$$d_1 = \text{numerator df}, \quad d_2 = \text{denominator df}$$
- **Definition:**
$$F = \frac{\chi_1^2/d_1}{\chi_2^2/d_2}$$
- **Mean:**
$$E(F) = \frac{d_2}{d_2 - 2}, \quad d_2 > 2$$

Use case in AI/ML

The F distribution is used for comparing variances.
Applications:
- ANOVA
- Model comparison
- Regression significance testing

In machine learning, F-tests help determine whether additional features significantly improve model performance.